```c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <time.h>
#include <string.h>

#define FILE_SIZE (10 * 1024 * 1024) // 10 MB file for benchma>

// Traditional read/write method
void benchmark_traditional(const char *filename) {
    int fd = open(filename, O_RDWR);
    if (fd < 0) { perror("open failed"); return; }

    char *buffer = malloc(4096);
    off_t offset = 0;

    clock_t start = clock();
    while (offset < FILE_SIZE) {
        read(fd, buffer, 4096);
        // modify buffer slightly to simulate work
        buffer[0] = 'X';
        lseek(fd, offset, SEEK_SET);
        write(fd, buffer, 4096);
        offset += 4096;
    }
    clock_t end = clock();

    free(buffer);
    close(fd);
    printf("Traditional read/write time : %.6f seconds\n", (do>
```

```c
// Memory-mapped I/O (mmap) method
void benchmark_mmap(const char *filename) {
    int fd = open(filename, O_RDWR);
    if (fd < 0) { perror("open failed"); return; }

    // Map file into memory
    char *map = mmap(NULL, FILE_SIZE, PROT_READ | PROT_WRITE, >
    if (map == MAP_FAILED) {
        perror("mmap failed");
        close(fd);
        return;
    }

    clock_t start = clock();
    // Directly manipulate memory like a normal array (kernel >
    for (size_t i = 0; i < FILE_SIZE; i += 4096) {
        map[i] = 'Y';
    }
    // Ensure changes are written back to disk
    msync(map, FILE_SIZE, MS_SYNC);
    clock_t end = clock();

    munmap(map, FILE_SIZE);
    close(fd);
    printf("Memory-mapped I/O (mmap) time : %.6f seconds\n", (>
}

int main() {
    const char *test_file = "perf_test.dat";

    // Create a 10MB dummy file
    int fd = open(test_file, O_CREAT | O_RDWR | O_TRUNC, 0644);
    if (fd < 0) { perror("Failed to create test file"); exit(1>
```

```
gireesh@Kannigantiii:~/oslab$ echo "Hello Inode Lab" > original
.txt
ls -i original.txt
44861 original.txt
gireesh@Kannigantiii:~/oslab$ stat original.txt
  File: original.txt
  size: 16              Blocks: 8          IO Block: 4096   reg
ular file
Device: 8,48    Inode: 44861       Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ gireesh)   Gid: ( 1000/
 gireesh)
Access: 2026-10-27 12:53:49.454328868 +0000
Modify: 2026-10-27 12:53:49.454328868 +0000
Change: 2026-10-27 12:53:49.454328868 +0000
 Birth: 2026-10-27 12:53:49.454328868 +0000
gireesh@Kannigantiii:~/oslab$ ln original.txt hardlink.txt
ls -li original.txt hardlink.txt
44861 -rw-r--r-- 2 gireesh gireesh 16 Oct 27 12:53 hardlink.txt
44861 -rw-r--r-- 2 gireesh gireesh 16 Oct 27 12:53 original.txt
gireesh@Kannigantiii:~/oslab$ ln -s original.txt symlink.txt
ls -li original.txt hardlink.txt symlink.txt
44861 -rw-r--r-- 2 gireesh gireesh 16 Oct 27 12:53 hardlink.txt
44861 -rw-r--r-- 2 gireesh gireesh 16 Oct 27 12:53 original.txt
44926 lrwxrwxrwx 1 gireesh gireesh 12 Oct 27 12:54 symlink.txt
-> original.txt
gireesh@Kannigantiii:~/oslab$ # Get the inode number of origina
l.txt first, suppose it is 123456, then run:
find . -inum <inode_number>
-bash: syntax error near unexpected token `newline'
gireesh@Kannigantiii:~/oslab$ nano mmap_compare.c
gireesh@Kannigantiii:~/oslab$ gcc mmap_compare.c -o mmap_compar
e
./mmap_compare
=== Comparing Traditional I/O vs mmap() (10 MB File) ===
Traditional read/write time : 0.023316 seconds
Memory-mapped I/O (mmap) time : 0.002665 seconds
```

```
gireesh@Kannigantiii:~/oslab$ gcc mmap_compare.c -o mmap_compar
e
./mmap_compare
=== Comparing Traditional I/O vs mmap() (10 MB File) ===
Traditional read/write time : 0.023316 seconds
Memory-mapped I/O (mmap) time : 0.002665 seconds
```